

# 3. Model Selection and Architecture

ComicCraft Project Documentation

**Purpose:** Documents the model choices and the three-part architecture described in the source project.

## 3.1 Project Requirements

- Generate structured comic outlines panel-by-panel.
- Generate complete narration and character dialogues.
- Generate visually rich comic-style illustrations.
- Support customized prompts, characters, settings, tones, and art styles.

## 3.2 Model Selection

| Component                       | Selected model / technology                       |
|---------------------------------|---------------------------------------------------|
| Structured outline              | Gemini Flash (models/gemini-1.5-flash)            |
| Detailed narration and dialogue | Gemini Pro (models/gemini-1.5-pro)                |
| Comic illustrations             | Stable Diffusion (runwayml/stable-diffusion-v1-5) |
| Image pipeline                  | Hugging Face Diffusers                            |
| Backend                         | FastAPI                                           |
| Frontend rendering              | HTML, CSS, Jinja2                                 |
| PDF export                      | FPDF                                              |

## 3.3 Reasons Given in the Source

- Gemini Flash is selected for fast, structured generation of comic panel outlines.
- Gemini Pro is selected for detailed and creative narration and dialogue.
- Stable Diffusion is selected for comic-style illustrations from image prompts.
- The models are evaluated using response quality, prompt adherence, detail, image speed and resolution, stability, documentation, and community support.

## 3.4 Three-Layer Architecture

### Frontend

HTML, CSS, and Jinja2 collect user inputs and display generated comic panels, text, and export status.

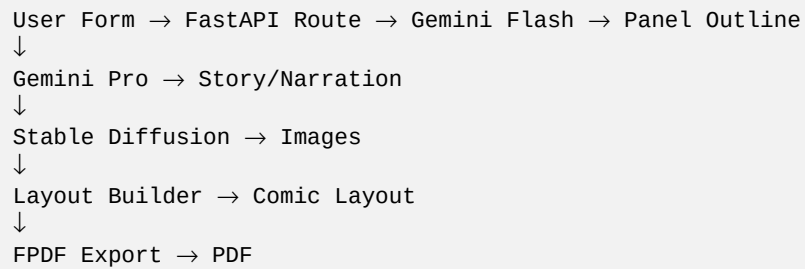
## Backend

FastAPI manages routes, form and JSON input, AI workflow coordination, layout construction, and PDF export.

## AI Integration

Gemini APIs generate story content while Hugging Face Diffusers and Stable Diffusion generate panel illustrations.

## 3.5 Data Flow



```
graph TD; A[User Form] --> B[FastAPI Route]; B --> C[Gemini Flash]; C --> D[Panel Outline]; D --> E[Gemini Pro]; E --> F[Story/Narration]; F --> G[Stable Diffusion]; G --> H[Images]; H --> I[Layout Builder]; I --> J[Comic Layout]; J --> K[FPDF Export]; K --> L[PDF];
```

User Form → FastAPI Route → Gemini Flash → Panel Outline  
↓  
Gemini Pro → Story/Narration  
↓  
Stable Diffusion → Images  
↓  
Layout Builder → Comic Layout  
↓  
FPDF Export → PDF